

Towards Text-Guided Navigation World Models

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Abstract

Navigation World Models (NWM) have recently attracted attention for reaching a goal location without a pre-built map. Using a diffusion model, an NWM imagines the scenes multiple candidate actions would produce and follows the trajectory whose predicted observation is closest to the goal image. Because selection rests entirely on this visual comparison, higher input resolution, which preserves finer visual detail, yields more accurate trajectories. Higher resolution, however, raises computation cost and navigation latency. A diffusion model repeats the denoising step hundreds of times, and even a modest increase in token count is amplified across every repetition, while attention memory grows accordingly. As Fig. 1 shows, raising the input resolution from 128 to 224 increases total runtime by $3.7\times$ and VRAM usage by 28%.

This paper proposes Text-Guided Navigation World Models (TG-NWM), which inject a textual description of the current observation alongside the image. This design aims to retain high navigation accuracy at low input resolution, enabling reliable navigation at reduced computational cost. Unlike Vision-Language Models (VLMs), which natively handle both text and images, a vision-only model such as NWM requires careful design to be conditioned on text. We present a prototype implementation of TG-NWM and discuss the challenges and design considerations that emerge from our preliminary results.

Fig. 4 shows the overall architecture of TG-NWM. The core module of NWM is the Conditional Diffusion Transformer (CDiT), which predicts the observation a candidate action would produce by iteratively denoising a latent from context frames. TG-NWM additionally supplies CDiT with a textual description of the most recently observed frame, generated with Qwen-VL from the captured image and encoded with CLIP, and inserts a cross-attention block to attend to the resulting text tokens. Because the CLIP text and VAE vision latent spaces were never jointly trained, we fully fine-tune the model on RECON such that generated frames better match the ground truth.

We evaluate TG-NWM from two perspectives: the fidelity of the imagined future observations, and the accuracy of the selected trajectory. Fig. 2 reports LPIPS between each predicted frame and the corresponding ground-truth frame, and Fig. 3 reports ATE, the deviation of the selected trajectory from the ground truth. The values are averaged across

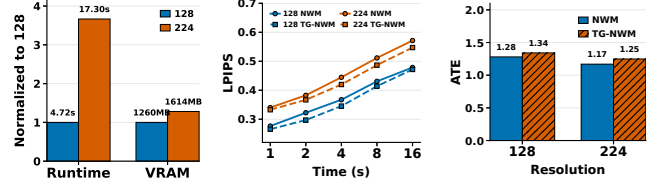


Figure 1: Cost

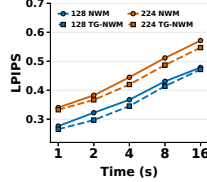


Figure 2: LPIPS

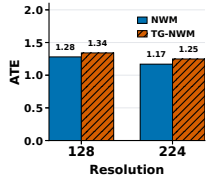


Figure 3: ATE

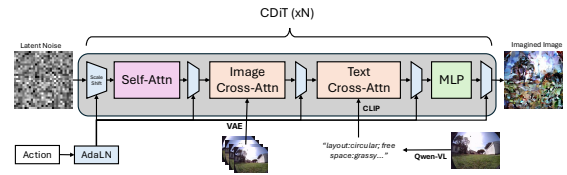


Figure 4: CDiT in TG-NWM

candidates and lower is better for both. The most prominent observation is that text conditioning lowers LPIPS on average (by 4.3%), yet trajectory accuracy degrades, with ATE increasing by 6.8%. We attribute this counter-intuitive result to a loss of sensitivity to the candidate action.

NWM samples a set of candidate trajectories (120 by default) and takes the mean displacement of the top- k ($k = 5$) as the selected trajectory. The textual description narrows the margin between candidates. Consequently, borderline, sub-optimal candidates can enter the elite set. Trajectory error increases in turn. In addition, fixed text conditioning throughout the rollout likely reinforces this effect. Our prototype currently reuses the text embedding of the initial observation at every prediction step. Visual frames, however, are updated throughout the rollout. This limitation might blur the difference between candidates, even as that difference would normally widen as trajectories progress.

Based on these observations, we plan two directions for future work. First, we will study methods that complement LPIPS to select an elite set that actually improves ATE. Second, we will implement continuous text updates throughout the rollout and examine its effectiveness and cost.

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